

STATS 506 - Homework 1

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[GitHub Repository] <https://github.com/bananajoemama/STATS506>

Problem 1 - Exploratory Analysis of Soybean Cultivars

Problem 1 Code

```
library(readr)
```

```
Warning: package 'readr' was built under R version 4.5.2
```

```
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.5.2
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```
library(knitr)
```

```
Warning: package 'knitr' was built under R version 4.5.2
```

```
# import data
soybean <- read_csv("data.csv", show_col_types = FALSE)

# verif and season counts
cat("Total Observations:", nrow(soybean), "\n")
```

```
Total Observations: 320
```

```
cat("Distinct Cultivars:", n_distinct(soybean$Cultivar), "\n")
```

```
Distinct Cultivars: 40
```

```
cat("Distinct Seasons:", n_distinct(soybean$Season), "\n")
```

```
Distinct Seasons: 2
```

```
# table of obs per season
table(soybean$Season)
```

```
 1  2
160 160
```

```
# overall corr between PH and GY
cor(soybean$PH, soybean$GY, use = "complete.obs")
```

```
[1] 0.1232807
```

```
# corr between PH and GY by season
soybean %>%
  group_by(Season) %>%
  summarize(correlation = cor(PH, GY, use = "complete.obs"))
```

```
# A tibble: 2 × 2
  Season correlation
  <dbl>         <dbl>
1     1         0.231
2     2        -0.0227
```

```
# yield of obs with greatest PH
soybean %>%
  filter(PH == max(PH, na.rm = TRUE)) %>%
  select(PH, GY)
```

```
# A tibble: 1 × 2
  PH     GY
<dbl> <dbl>
1  94.8 3305.
```

```
# % of obs with MHG > average MHG
mean(soybean$MHG > mean(soybean$MHG, na.rm = TRUE), na.rm = TRUE) * 100
```

```
[1] 47.1875
```

```
# cultivar with highest average grain yield
soybean %>%
  group_by(Cultivar) %>%
  summarize(avg_GY = mean(GY, na.rm = TRUE)) %>%
  arrange(desc(avg_GY)) %>%
  slice(1)
```

```
# A tibble: 1 × 2
  Cultivar avg_GY
<chr>      <dbl>
1 98R30 CE  3965.
```

```
# quantitative var summary table
quant_vars <- c("PH", "IFP", "NLP", "NGP", "NGL", "NS", "MHG", "GY")

overall_avg <- soybean %>%
  summarize(across(all_of(quant_vars), ~ mean(.x, na.rm = TRUE))) %>%
  mutate(Group = "Overall")

season_avg <- soybean %>%
  group_by(Season) %>%
  summarize(across(all_of(quant_vars), ~ mean(.x, na.rm = TRUE))) %>%
  mutate(Group = paste("Season", Season)) %>%
  select(-Season)

bind_rows(overall_avg, season_avg) %>%
  select(Group, everything()) %>%
  kable(digits = 2, caption = "Average Values of Quantitative Variables")
```

Group	PH	IFP	NLP	NGP	NGL	NS	MHG	GY
Overall	68.39	15.46	59.09	135.09	2.29	4.07	168.32	3418.55
Season 1	65.99	16.48	57.97	135.24	2.30	4.82	162.29	3428.45
Season 2	70.79	14.45	60.21	134.94	2.28	3.33	174.36	3408.66

```
# 2-sample t-test
t.test(GY ~ Season, data = soybean)
```

Welch Two Sample t-test

```
data: GY by Season
t = 0.35155, df = 231.85, p-value = 0.7255
alternative hypothesis: true difference in means between group 1 and group 2
is not equal to 0
95 percent confidence interval:
-91.15712 130.75170
sample estimates:
mean in group 1 mean in group 2
3428.452 3408.655
```

```
# manual t-test calculation
s1 <- soybean %>% filter(Season == 1) %>% pull(GY)
s2 <- soybean %>% filter(Season == 2) %>% pull(GY)

n1 <- length(s1); n2 <- length(s2)
m1 <- mean(s1); m2 <- mean(s2)
v1 <- var(s1); v2 <- var(s2)

t_stat <- (m1 - m2) / sqrt((v1/n1) + (v2/n2))
df_welch <- ((v1/n1) + (v2/n2))^2 / (((v1/n1)^2 / (n1 - 1)) + ((v2/n2)^2 / (n2 - 1)))
p_val <- 2 * pt(-abs(t_stat), df = df_welch)

cat("Manual t-stat:", t_stat, "\nManual df:", df_welch, "\nManual p-value:",
p_val, "\n")
```

```
Manual t-stat: 0.351546
Manual df: 231.8466
Manual p-value: 0.725498
```

Problem 1 Written Answers

Dataset Verification

Observations: 320 total observations across 40 distinct cultivars and 2 seasons.

Seasonal Distribution: Season 1 contains 160 observations; Season 2 contains 160 observations.

Exploratory Data Analysis

Overall Correlation: The correlation between plant height (PH) and grain yield (GY) across all observations is 0.1233.

Correlation by Season: Season 1 has a higher correlation (0.2309) compared to Season 2 (-0.0227).

Yield at Maximum Height: The observation with the maximum plant height (94.80 cm) produced a grain yield of 3304.68.

Above-Average Seed Weight: 47.19% of observations have a thousand seed weight (MHG) greater than the overall mean MHG.

Top Performing Cultivar: 98R30 CE achieved the highest overall average grain yield (3964.97).

Summary Table

Group	PH	IFP	NLP	NGP	NGL	NS	MHG	GY
Overall	68.39	15.47	59.09	135.09	2.29	4.07	168.32	3418.55
Season 1	65.99	16.49	57.97	135.24	2.30	4.82	162.29	3428.45
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Two-Sample t-Test on Grain Yield Across Seasons

Null Hypothesis H_0 : $\mu_1 = \mu_2$ (Mean grain yield in Season 1 is equal to Season 2).

Alternative Hypothesis (H_a): $\mu_1 \neq \mu_2$ (Mean grain yield in Season 1 is not equal to Season 2).

Test Results: Welch two-sample t -test statistic $t = 0.3515$, $df = 231.85$, p -value = 0.7255. Manual calculation confirms $t = 0.3515$, $df = 231.85$, p -value = 0.7255.

Interpretation: Because $p = 0.7255 > 0.05$, we fail to reject the null hypothesis at the $\alpha = 0.05$ significance level. There is no statistically significant difference in mean grain yield between Season 1 and Season 2.

Problem 2

Problem 2 Code

```
library(readr)
library(dplyr)
library(knitr)
```

```

# import data
col_names <- c("age", "workclass", "fnlwgt", "education", "education_num",
               "marital_status", "occupation", "relationship", "race", "sex",
               "capital_gain", "capital_loss", "hours_per_week",
               "native_country", "income")

adult <- read_csv("adult.data", col_names = col_names, na = c("?", " ?"),
trim_ws = TRUE, show_col_types = FALSE)

# examine missing values per var
missing_summary <- adult %>%
  summarize(across(everything(), ~ sum(is.na(.x)))) %>%
  tidyr::pivot_longer(cols = everything(), names_to = "var", values_to =
"missing_count")

missing_summary %>%
  kable(caption = "missing obs per var")

```

var	missing_count
age	0
workclass	1836
fnlwgt	0
education	0
education_num	0
marital_status	0
occupation	1843
relationship	0
race	0
sex	0
capital_gain	0
capital_loss	0
hours_per_week	0
native_country	583
income	0

```

# sample size before filtering
n_initial <- nrow(adult)
cat("initial sample size:", n_initial, "\n")

```

```
initial sample size: 32561
```

```
# filter hours worked >= 10 and confirm sample size
adult_step1 <- adult %>%
  filter(hours_per_week >= 10)

n_step1 <- nrow(adult_step1)
cat("sample size hours worked >= 10:", n_step1, "\n")
```

```
sample size hours worked >= 10: 32103
```

```
# remove missing workclass occupation native country
adult_step2 <- adult_step1 %>%
  filter(!is.na(workclass) & !is.na(occupation) & !is.na(native_country))

n_step2 <- nrow(adult_step2)
cat("sample size after removing missing values:", n_step2, "\n")
```

```
sample size after removing missing values: 29873
```

```
# filter age and education
adult_final <- adult_step2 %>%
  filter(age >= 18, age <= 90, education != "Preschool")

# final sample size
n_final <- nrow(adult_final)
cat("final sample size:", n_final, "\n")
```

```
final sample size: 29523
```

Problem 2 Written Answers

Missing Value Examination

Missing values in the raw dataset are concentrated within three specific categorical variables:

workclass: 1,836 missing values

occupation: 1,843 missing values

native_country: 583 missing values

All other 12 variables: 0 missing values

Sample Filtering & Step-by-Step Sample Sizes

Initial Sample Size: 32,561 observations.

Hours Worked Restricting (`hours_per_week >= 10`): 32,103 observations remaining (458 observations removed for working under 10 hours per week).

Missing Value Removal (`workclass`, `occupation`, `native_country`): 29,873 observations remaining (2,230 incomplete observations removed).

Explanation for Removal: `workclass`, `occupation`, and `native_country` represent key demographic and structural employment variables. Imputing values for these categorical fields would introduce arbitrary classification bias. Dropping complete cases with missing entry labels ensures that subsequent statistical comparisons reflect genuine reported attributes.

Demographic & Education Exclusions (`18 <= age <= 90 & education != "Preschool"`): Excluded individuals under 18 (who fall outside standard adult workforce dynamics), individuals over 90 (to filter potential top-coded boundary artifacts), and respondents with a "Preschool" education level.

Final Sample Size

Final Cleaned Dataset: 29,523 observations.

Problem 3

Problem 3 Code

```
#' compute next value in happy sequence
#'
#' @param n a positive integer
#' @return sum of squared digits
nextHappy <- function(n) {
  # input validation
  if (!is.numeric(n) || length(n) != 1 || n <= 0 || n != floor(n)) {
    stop("input n must be a single positive integer")
  }

  # split digits and calculate sum of squares
  digits <- as.numeric(strsplit(as.character(n), "")[[1]])
  sum(digits^2)
}

# test examples
nextHappy(19)
```

```
[1] 82
```

```
nextHappy(82)
```

```
[1] 68
```

```
#' generate sequence of happy numbers
#
#' @param n a positive integer
#' @return list with sequence vector and isHappy boolean
happySequence <- function(n) {
  # input validation
  if (!is.numeric(n) || length(n) != 1 || n <= 0 || n != floor(n)) {
    stop("input n must be a single positive integer")
  }

  seq_vec <- n
  curr <- n

  # iteratively compute next sequence value
  while (TRUE) {
    curr <- nextHappy(curr)

    if (curr == 1) {
      seq_vec <- c(seq_vec, 1)
      return(list(sequence = seq_vec, isHappy = TRUE))
    }

    if (curr %in% seq_vec) {
      seq_vec <- c(seq_vec, curr)
      return(list(sequence = seq_vec, isHappy = FALSE))
    }

    seq_vec <- c(seq_vec, curr)
  }
}

# test examples
happySequence(19)
```

```
$sequence
[1] 19 82 68 100 1

$isHappy
[1] TRUE
```

```
happySequence(4)$isHappy
```

```
[1] FALSE
```

```
# run happySequence across 1:1000
results <- lapply(1:1000, happySequence)

# count and percentage of happy numbers
is_happy <- sapply(results, function(x) x$isHappy)
num_happy <- sum(is_happy)
pct_happy <- (num_happy / 1000) * 100

# sequence lengths
seq_lens <- sapply(results, function(x) length(x$sequence))

# overall longest sequence length
max_len_all <- max(seq_lens)
starters_all <- which(seq_lens == max_len_all)

# longest sequence among happy numbers only
happy_lens <- seq_lens
happy_lens[!is_happy] <- 0
max_len_happy <- max(happy_lens)
starters_happy <- which(happy_lens == max_len_happy)

cat("number of happy numbers:", num_happy, "\n")
```

```
number of happy numbers: 143
```

```
cat("percentage of happy numbers:", pct_happy, "%\n")
```

```
percentage of happy numbers: 14.3 %
```

```
cat("smallest starting value for longest overall sequence:",
min(starters_all), "\n")
```

```
smallest starting value for longest overall sequence: 269
```

```
cat("smallest starting value for longest happy sequence:",
min(starters_happy), "\n")
```

```
smallest starting value for longest happy sequence: 356
```

Problem 3 Written Answers

Number of Happy Numbers: 143

Percentage of Happy Numbers: 14.3%

Longest Sequence Starting Value (All Inputs): 269 produces the longest overall sequence with 20 steps before entering a cycle (tied with 296, 629, 667, 676, 692, 766, 799, 926, 962, 979, and 997).

Longest Sequence Starting Value (Happy Numbers Only): 356 produces the longest sequence among terminating happy numbers with 7 steps (tied with 365, 379, 397, 469, 496, 536, 563, 635, 649, 653, 694, 739, 793, 899, 937, 946, 964, 973, 989, and 998).